

1. $f: (0,1) \rightarrow \mathbb{R}$

$f(x) = 1$, if x is irrational, $x \in (0,1)$
 $= -1$ if x is rational, $x \in (0,1)$

i) $\lim_{x \rightarrow a} f(x)$ does not exist $\forall a \in [0,1]$

ii) $|f|$ is continuous but f is not continuous

iii) f is not R-integrable

2. $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$f(x) = 1, x \in \mathbb{Q}$

$= 0, x \in \mathbb{R} - \mathbb{Q}$

i) f is continuous at no point $c \in \mathbb{R}$.

ii) f is not R-integrable.

3. $f: [0,1] \rightarrow \mathbb{R}$ by

$f(x) = x$, x is rational in $[0,1]$

$= 1-x$, x is irrational in $[0,1]$

i) f is injective on $[0,1]$

ii) f assumes every real no. in $[0,1]$

iii) f is continuous at $\frac{1}{2}$ & discontinuous at every point in $[0,1]$

iv) f is not R-integrable.

4. $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$f(x) = x, x \in \mathbb{Q}$

$= 0, x \in \mathbb{R} - \mathbb{Q}$

i) f is continuous at 0 and every other point x has a discontinuity of second kind.

ii) f is not R-integrable.

5. $f: [0,1] \rightarrow \mathbb{R}$ by

$f(x) = 0$, if x is irrational

$= \frac{1}{n}$, if $x = \frac{m}{n}$, where $\gcd(m,n)=1$

i) f is continuous at every irrational pt & discontinuous at every rational point.

ii) f is not differentiable at 0.

iii) f is R-integrable on $[0,1]$ & $\int_0^1 f = 0$

6. $f: [0,1] \rightarrow \mathbb{R}$ by

$f(x) = x$ if x is rational

$= x^2$ if x is irrational

i) f is not R-integrable.

7. $f: [0,1] \rightarrow \mathbb{R}$ by

$f(x) = \sqrt{1-x^2}, x \in [0,1] \cap \mathbb{Q}$

$= 1-x, x \in [0,1] - \mathbb{Q}$

i) f is not R-integrable.

8. $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$f(x) = \frac{1}{x^2}$ if x is rational > 1

$= -\frac{1}{x^2}$ if x is irrational > 1

i) $\int_1^\infty |f(x)| dx$ convergent.

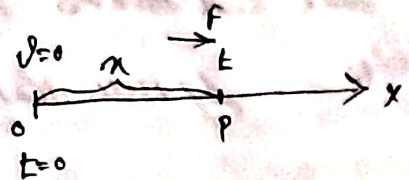
9. $\frac{\partial}{\partial z} \int_{a(z)}^{b(z)} f(x,z) dx = \int_{a(z)}^{b(z)} \frac{\partial f}{\partial z} dx + f(b(z), z) \frac{\partial b}{\partial z} - f(a(z), z) \frac{\partial a}{\partial z}$

§ Dynamics of a particle §

Motion in a 1D line

§ Equation of motion:-

Let O be the origin and a particle start from O and move along OX . Let P be the position of the particle at any time t , such that $OP = x$. Let v be the velocity of the particle at P . Then $v = \frac{dx}{dt}$ = Velocity of the particle at P [in the direction of x -axis]



$$\begin{aligned}
 \text{Now } f &= \frac{dv}{dt} = \text{Acceleration of the particle at } P \\
 &= \frac{dv}{dt} \text{ (in the direction of } x \text{ increasing)} \\
 &= v \cdot \frac{dv}{dx}
 \end{aligned}$$

Then equation of motion of the particle of mass m is

$$m \ddot{x} = F \quad [F \text{ being the external force acting on the particle in the positive direction of } x\text{-axis}]$$

$$\text{or } m \ddot{x} = -F \quad [F \text{ being the external force acting on the particle in negative direction of } x\text{-axis}]$$

§ Important Relations:-

1. $v = u + ft$, u being the initial velocity.
2. $x = ut + \frac{1}{2}ft^2$
3. $v^2 = u^2 + 2fx$

§ Prob. 1:- If the Velocity v of a particle moving in a str. line is given by $v = ax + b$, where x is the distance travelled from a fixed point on the line and a, b are constants, the acceleration varies as —

Ans.

Given that $v = ax + b$
 D.W. r. to x
 $\frac{dv}{dx} = a$

$\therefore a = \frac{dv}{dx}$ [as $v \frac{dt}{dx} = a$ = acceleration]
 a for x , a being the constant.

§ Prob. 2:- A particle moves along the str. line according to the law $s = 6t^2 + 4t + 3$, where s is the distance of the particle at time t . Then the Velocity of the particle at time $t = 1$ is equal to —

Ans.

Given that $s = 6t^2 + 4t + 3$

D.W. r. to t

$\frac{ds}{dt} = 12t + 4$

$\therefore \frac{ds}{dt} = \frac{12t + 4}{\sqrt{6t^2 + 4t + 3}}$

$\therefore \left. \frac{ds}{dt} \right|_{t=1} = \frac{12 \cdot 1 + 4}{\sqrt{6 \cdot 1^2 + 4 \cdot 1 + 3}} = \frac{16}{\sqrt{13}}$

$\therefore \left. \frac{ds}{dt} \right|_{t=1} = \frac{16}{\sqrt{13}}$

§ Prob. 3:- The Velocity of a particle moving in a str. line given by the relation $v = ax^2 + bx + c$, where v is the Velocity of the particle and x is the distance of the particle from fixed point at time t . The acceleration of the particle is —

Ans.

Given that $v = ax^2 + bx + c$

D.W. r. to x

$\frac{dv}{dx} = 2ax + b$

$\therefore \left[\frac{dv}{dx} = 2ax + b \right]$

§ Prob: 4:- Which of the following is scalar?

i) Displacement ii) Speed iii) Velocity iv) Acceleration

Ans. Speed is a scalar quantity. It has magnitude but does not have direction.

§ Prob: 5:- A particle moves along a st. line according to the law $S = 6t^3 + 4t + 3$, where S is the displacement and t time, then its acceleration varies as —

Ans. Given that $S = 6t^3 + 4t + 3$

∴ D. w. r. to t

$$\frac{dS}{dt} = 18t^2 + 4$$

$$\text{or } \frac{d^2S}{dt^2} = 36t$$

$$\text{or } \frac{d^2S}{dt^2} \propto t$$

$$\text{or } \frac{d^2S}{dt^2} \propto \frac{1}{t^2}$$

$$\text{or } \frac{d^2S}{dt^2} = \frac{36t^3 - 36t^2 - 24t - 4}{t^3}$$

$$= \frac{36t^3 + 24t + 18 - 36t^2 - 24t - 4}{t^3}$$

$$\text{or } \frac{d^2S}{dt^2} = \frac{14}{t^3}$$

$$\text{or } \frac{d^2S}{dt^2} \propto \frac{1}{t^3}$$

$$\text{or } \frac{d^2S}{dt^2} \propto \frac{1}{t^3}$$

§ Prob: 5:- The law of motion of a particle moving in a st. line is $S = \frac{1}{2}vt^2$. Then the acceleration is —

Ans. Given that $S = \frac{1}{2}vt^2$

∴ D. w. r. to t

$$\frac{dS}{dt} = \frac{1}{2} [v \cdot \frac{d}{dt} t^2 + t^2 \cdot \frac{dv}{dt}]$$

$$\text{or } 2v = \frac{dv}{dt} \cdot t$$

$$\text{or } v = f \cdot t$$

$$\text{S.D. } a = \frac{dv}{dt}$$

$$\frac{dv}{dt} = f \cdot 1 + t \cdot \frac{df}{dt}$$

$$\text{or } t \cdot \frac{df}{dt} = 0$$

$$\Rightarrow t = 0 \quad / \quad \frac{df}{dt} = 0$$

which is impossible. $\Rightarrow f$ is constant.

§ Prob. 6:- If the time t be regarded as function of velocity v , then, the rate of decrease of acceleration a is —

Ans. Given that $t = g(v)$

$$\text{Now } \frac{dt}{dv} = \frac{\frac{dt}{dv}}{\frac{dv}{dt}} \quad [\text{By chain rule}]$$

$$= \frac{d}{dv} \left(\frac{dt}{dv} \right) \cdot \frac{dv}{dt}$$

$$= \frac{d}{dv} \left(\frac{dt}{dv} \right) \cdot f$$

$$= \frac{d}{dv} \left(\frac{1}{\frac{dv}{dt}} \right) \cdot f$$

$$= f \cdot \left(-\frac{1}{\left(\frac{dv}{dt} \right)^2} \right) \cdot \frac{dv}{dt}$$

$$\boxed{\frac{dt}{dv} = -f \cdot \frac{dv}{dt^2}}$$

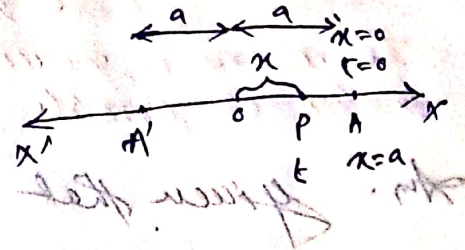
§ Prob. 7:- A point moves in a st. line so that its distance s from a fixed point at time t is proportional to t^n . If v be the velocity & a the acceleration at any time t , show that $\boxed{(n-1)v = na}$

Simple Harmonic Motion

§ Definition:- A particle is said to execute a simple harmonic motion (S.H.M.), if it moves in a str. line such that its acceleration is always directed towards a fixed point in the line and is proportional to its distance from the fixed point.

§ Equation of motion of S.H.M.:-

$$\begin{aligned} \ddot{x} &= -\mu x \\ \Rightarrow \int \ddot{x} &= \int -\mu x \\ \Rightarrow \dot{x} &= -\mu x \end{aligned}$$



$$\text{Periodic Time (T)} = \frac{2\pi}{\sqrt{\mu}}$$

$$\text{Frequency of oscillation (n)} = \frac{1}{T} = \frac{\sqrt{\mu}}{2\pi}$$

$$\text{Amplitude of S.H.M.} = a$$

§ General form:-

$$x = c \cos(\sqrt{\mu}t + \epsilon)$$

The quantity ϵ is called epoch and $(\sqrt{\mu}t + \epsilon)$ is called argument of the S.H.M.

$$\text{Phase time} = t + \frac{\epsilon}{\sqrt{\mu}}$$

Ex Problem 1:- The speed v of a point moving along x -axis is given by $v = 16 - x$. Prove that the motion is simple harmonic. Find also amplitude.

Ans. Given that $v = 16 - x$

D.D. $\sim$ to x

$$Q.D. \frac{dv}{dx} = -1$$

$$a \ddot{x} = -x$$

which indicates that the motion of the particle is Simple Harmonic motion.

Again

$$v = 16 - x$$

Putting $v = 0$

$$16 - x = 0$$

$$\Rightarrow x = 16$$

$$\therefore \text{So amplitude} = \frac{16 - (-16)}{2} = 16$$

Problem 2:- If the velocity of a particle moving along the x -axis is given by $v = 86x - x^2 - 126$. Show that the motion is simple harmonic motion.

Ans. Given that $v = 86x - x^2 - 126$
D. w. r. to x
 $20 \frac{dv}{dx} = 86 - 2x$
 $\therefore \ddot{x} = -(12 - 46)$

Which indicates that the motion of the particle is a simple harmonic motion about the mean position $x = 46$.

Problem 3:- A particle moves along a st. line under the law of force given by $x \sin(\pi t + b)$. Show that the acceleration is directed towards the origin and varies as the distance.

Ans. Given that $x = a \cos(\pi t + b)$
D. w. r. to t
 $\frac{dx}{dt} = -a\pi \sin(\pi t + b)$
D. w. r. to t
 $\ddot{x} = -a\pi^2 \cos(\pi t + b)$
 $\therefore \ddot{x} = -\pi^2 x$

which shows that the acceleration is directed towards the origin and varies as the distance.

Problem 4:- A particle describes simple harmonic motion in a line with 2 cm as its amplitude. Its velocity when passing through the centre of oscillation is 12 cm/sec. Find the period.

Ans. For a particle executing S.H.M., the displacement x at any time t is given by

$$x = a \cos \omega t$$

$$\text{S.D. } v = \dot{x}$$

$$\dot{x} = -a\omega \sin \omega t$$

$$= -a\omega \sqrt{1 - \cos^2 \omega t}$$

$$\dot{x} = -a\omega \sqrt{1 - \frac{x^2}{a^2}}$$

Here amplitude $(a) = 2 \text{ cm}$ and when $x = 0$, $v = 12 \text{ cm/s}$

$$1. \quad |\dot{x}| = \sqrt{12^2 - 0^2} = 12$$

$$a \omega = 12$$

$$(2 \times \omega) \omega = 6 \Rightarrow \omega = 3$$

$$\text{Period} = \frac{2\pi}{\omega} = \frac{2\pi}{3} = \frac{2\pi}{3} \text{ s}$$

§ Problem 5:- The distances of a particle, performing simple harmonic motion, from the middle point of its path at three consecutive seconds are observed to be x, y, z . Show that the time of a complete oscillation is $\frac{2\pi}{\cos^{-1}(\frac{x+z}{2y})}$.

Ans. Let the distances of the particle from the middle point of its path at $(t-1)$, t and $(t+1)$ be x, y and z respectively.

Now equation of motion of the particle is

$$\ddot{x} = -\mu x$$

and a be the amplitude of the motion.

$$\text{Then we have } x = a \cos \omega(t-1) \quad \text{--- (i)}$$

$$y = a \cos \omega t \quad \text{--- (ii)}$$

$$z = a \cos \omega(t+1) \quad \text{--- (iii)}$$

$$\text{Now } x+z = a \cos \omega(t-1) + a \cos \omega(t+1)$$

$$= 2a \cos \omega t \cdot \cos \omega$$

$$= 2y \cdot \cos \omega$$

$$\Rightarrow \cos \omega = \frac{x+z}{2y} \Rightarrow \omega = \cos^{-1}\left(\frac{x+z}{2y}\right)$$

$\therefore$ Time of a complete oscillation = Periodic time

$$T = \frac{2\pi}{\sqrt{\mu}} = \frac{2\pi}{\cos\left(\frac{\pi+2}{2\pi}\right)}$$

Problem 6:- In a simple harmonic motion, if f be the acceleration and v the velocity at any instant and T is the periodic time, then show that $f^2 T^2 + 4\pi^2 v^2$ is constant.

Ans. Let the equation be

$$\ddot{x} = -\mu x$$

and a be the amplitude of the motion.

$$\text{Then } f = -\mu x, T = \frac{2\pi}{\sqrt{\mu}} \text{ and } v = \mu(a^2 - x^2)$$

$$\text{Now } f^2 T^2 + 4\pi^2 v^2 = \mu^2 x^2 \cdot \frac{4\pi^2}{\mu} + 4\pi^2 \mu (a^2 - x^2) \\ = 4\pi^2 a^2 \mu \text{ which is constant.}$$

Problem 7:- A particle of mass m moves along a straight line under an attraction $m n^2 x$ towards a point O on the line, where x is the distance from O . If $x = a$ and $\dot{x} = u$, when $t = 0$, then show that the amplitude of motion is $\sqrt{a^2 + \frac{u^2}{n^2}}$.

In particular, if $u = 0$, then find the period of oscillation.

Ans. The equation of motion is

$$m \ddot{x} = -m n^2 x$$

$$\text{or } \ddot{x} = -n^2 x \quad \text{--- (1)}$$

Multiplying both sides by $2\dot{x}$

$$2\dot{x} \ddot{x} = -n^2 x 2\dot{x}$$

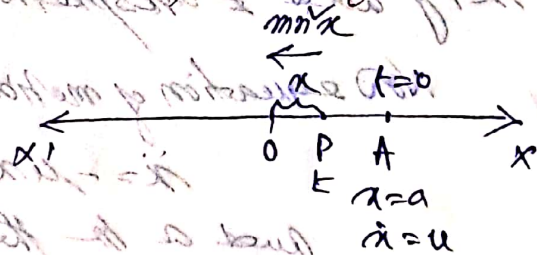
$$\text{or } \frac{d}{dt}(\dot{x}^2) = -2n^2 x \frac{dx}{dt}$$

Integrating w.r. to t

$$\int \frac{d}{dt}(\dot{x}^2) dt = -2n^2 \int x \frac{dx}{dt} dt$$

$$\text{or } \int d(\dot{x}^2) = -2n^2 \int x dx$$

$$\text{or } \dot{x}^2 = -2n^2 x^2/2 + C, C = \text{const.}$$



§ For S.H.M

1. Equation of motion

$$\ddot{x} \propto v \frac{dv}{dx} = -\mu x (\mu > 0)$$

2. Velocity: $v = -\sqrt{\mu(a^2 - x^2)}$

3. Displacement:

$$x = a \cos(\sqrt{\mu}t + \epsilon)$$

$$x = A \cos \sqrt{\mu}t + B \sin \sqrt{\mu}t$$

4. Period: $T = \frac{2\pi}{\sqrt{\mu}}$

5. Amplitude = a

6. Epoch = ϵ , Phase = $\sqrt{\mu}t + \epsilon$

Phase time = $t + \epsilon/\sqrt{\mu}$

For damped oscillation

Equation of motion:

$$\ddot{x} + \mu \dot{x} + n^2 x = 0$$

Displacement:

$$x = c e^{-\frac{\mu}{2n}t} \cos(pt + \epsilon)$$

where $p^2 = n^2 - \mu^2/4$

For forced oscillation

$$\ddot{x} + n^2 x = F \cos pt$$

Displacement:

$$x = a \cos(nt + \epsilon) + \frac{F}{n^2 - p^2} \cos pt$$

For damped forced oscillation

Equation of motion:

$$\ddot{x} + Q\mu \dot{x} + n^2 x = K \cos pt$$

Displacement:

$$x = A e^{-\frac{Q\mu}{2n}t} \cos(\sqrt{n^2 - \frac{Q^2\mu^2}{4}}t + \epsilon) + \frac{K}{\sqrt{(n^2 - \frac{Q^2\mu^2}{4})^2 + (Q\mu p)^2}} \cos(pt + \epsilon_1)$$

$$\epsilon_1 = \tan^{-1} \left(\frac{2Q\mu p}{n^2 - \frac{Q^2\mu^2}{4}} \right)$$



Impulse and Impulsive Force

§ Impulse of a force:-

The impulse of a force is defined to be product of the force and time during which the force acts on a mass, when the force is constant in magnitude and direction.

II 1. $\boxed{\text{Impulse (I)} = P \times t}$, $P = \text{Force}$ and $t = \text{time duration}$

2. $\text{Impulse (I)} = m(v - u)$

$\boxed{\text{Impulse} = \text{change in momentum}}$

3. $\boxed{\text{Work done} = I \cdot \frac{1}{2}(u+v)}$

§ Impulsive force:- An impulsive force is a very large force of an extremely short duration, such that impulse of the force or the change in momentum produced by it in the body, is finite, but the displacement of the body during the short interval is negligible.

§ Principle of Conservation of Linear momentum:-

If the sum of the external forces acting on a system of bodies be zero in any direction, then the total momentum of the system in that direction remains constant during the motion.

$$\boxed{mu + mV = m'v + M'V}$$

§ For gun recoil:- $\boxed{M'V = m'v}$

§ Problem 1:- A shot of mass 40 kg leaves a gun with a velocity of 1000 cm/sec. If the velocity of recoil is 10 cm/sec. Find the mass of the gun.

Ans. Let the mass of the gun be M kg.

By the principle of linear momentum

$$M \times 10 \times 1000 = 40 \times 1000 \times 1000$$

$$\text{or } M = 4000 \text{ kg.}$$

§ Problem 2:- A force acting on a body of mass 5 lb changes its velocity from 30 m.p.h. to 45 m.p.h. Find the impulse of the force.

Ans. Impulse of the force (D) = $m(v - u)$

$$= 5 \times \frac{(45 - 30)}{60 \times 60} \times 1760 \times 3 \text{ poundal sec.}$$

$$= 110 \text{ poundal sec.}$$

§ Problem 3:- A mass of 5 lbs attached to a string is at rest. If a sudden jerk of the string produces in the mass, a velocity of 12 ft/sec. Find the impulse of the jerk.

Ans. Impulse of the jerk

= Change in momentum

$$= m(v - u)$$

$$= 5(12 - 0) \text{ Poundal sec.}$$

$$= 60 \text{ Poundal sec.}$$

§ Problem 4:- When the gun powder bursts inside a gun of mass 10 lbs, the bullet of mass 1 ounce is projected with a velocity of 1000 ft/sec. What is the kinetic energy generated by the explosion.

Ans. Here, mass of the bullet = 1 ounce = $\frac{1}{16}$ lbs.

If v be the velocity of recoil of the gun.

Then by the principle of conservation

$$10v = \frac{1}{16} \times 1000$$

$$v = \frac{25}{4} \text{ ft/sec.}$$

If the total kinetic energy generated.

$$E = \text{K.E. of the gun} + \text{K.E. of the bullet.}$$

$$= \frac{1}{2} \times 10 \times \left(\frac{25}{4}\right)^2 + \frac{1}{2} \times \frac{1}{16} \times (1000)^2 \text{ ft Poundal}$$

§ Problem 5:- A cricket ball moving with a velocity of 25 m/s is struck by a bat which causes it to move in the same straight line but in opposite direction with a velocity of 15 m/s . If the impulse generated by the bat is 600 N-s . Find the mass of the ball.

Ans. Here $u = 25 \text{ m/s} = 2500 \text{ cm/s}$
and $v = 15 \text{ m/s} = 1500 \text{ cm/s}$ but in the direction opposite to that of u .

Let m be the mass of the ball.

Here $I = 600 \text{ N-s}$.

$$\text{Now } I = m(v - (-u))$$

$$600 = m(1500 + 2500)$$

$$600 = m(4000)$$

$$m = \frac{600}{4000} = \frac{3}{20} \text{ gm.}$$

§ Problem 6:- A shell of mass m is projected from a gun of mass M by an explosion which generates kinetic energy E . Prove that the initial velocity of the shell is $\sqrt{\frac{2ME}{(M+m)m}}$.

Ans. Let v_1 be the initial velocity of the shell and v_2 be the velocity of mass M .
By the principle of conservation of linear momentum
$$mv_1 = Mv_2 \quad \text{--- (1)}$$

Now the KE generated in the system for gun as well as the shell together is

$$E = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2$$

$$E = \frac{1}{2}m\left[v_1 + \left(\frac{mv_1}{M}\right)\right]^2$$

$$E = \frac{1}{2}m\left[v_1^2 + \frac{2m}{M}v_1^2 + \frac{m^2}{M^2}v_1^2\right]$$

$$E = \frac{1}{2}m\left[v_1^2\left(1 + \frac{2m}{M} + \frac{m}{M}\right)\right]$$

$$E = \frac{1}{2}m\left[v_1^2\left(\frac{M+m}{M}\right)\right]$$

$$2E = \frac{m(M+m)}{M}v_1^2$$

$$v_1^2 = \frac{2EM}{m(M+m)}$$

$$v_1 = \sqrt{\frac{2EM}{m(M+m)}}$$

* $v_1 \leftarrow$

8 Problem 7: A body of mass $(m_1 + m_2)$ split into two parts of masses m_1 and m_2 by an internal explosion which generates kinetic energy E . Show that if after explosion the parts move in the same line as before, their relative speed is $\sqrt{\frac{2E(m_1 + m_2)}{m_1 m_2}}$.

Ans. Let v be the velocity of the body before explosion and v_1 & v_2 be the velocities of two parts of the body m_1 and m_2 after explosion respectively.

Now by the principle of conservation of linear momentum we get,

$$(m_1 + m_2)v = m_1 v_1 + m_2 v_2 \quad \text{--- (1)}$$

After explosion kinetic energy generated,

$$\therefore E = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \frac{1}{2} (m_1 + m_2) v^2$$

$$2E = m_1 v_1^2 + m_2 v_2^2 - (m_1 + m_2) \left\{ \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \right\}^2$$

$$\therefore 2E(m_1 + m_2) = (m_1 + m_2) (m_1 v_1^2 + m_2 v_2^2) - (m_1 v_1 + m_2 v_2)^2$$

$$= m_1 m_2 (v_1^2 + v_2^2 - 2v_1 v_2)$$

$$\therefore (v_1 - v_2)^2 = \frac{2E(m_1 + m_2)}{m_1 m_2}$$

$$\therefore v_1 - v_2 = \sqrt{\frac{2E(m_1 + m_2)}{m_1 m_2}}$$

9 Problem 8: A gun of mass M fires a shell of mass m horizontally and the energy of explosion is such as would be sufficient to project the shell vertically to a height h . Show that the velocity of recoil of the gun is $\left\{ \frac{2mgh}{M(m+M)} \right\}^{1/2}$.

Ans. Let E be the energy generated by the explosion.

Since the energy is just sufficient to raise a mass m to a height h .

$$E = mgh \quad \text{--- (1)}$$

If u be the velocity of the shell and v that of recoil of the gun.

$$mu = Mv$$

$$\therefore u = \frac{Mv}{m} \quad \text{--- (2)}$$

$$\text{Energy of explosion} = E = \frac{1}{2} m u^2 + \frac{1}{2} M v^2$$

$$\therefore mgh = \frac{1}{2} m \frac{M^2 v^2}{m^2} + \frac{1}{2} M v^2$$

§ Problem 11:- A shell, lying in a straight smooth horizontal tube, suddenly explodes and breaks into portions of masses m_1 and m_2 . If s is the distance apart of the masses after a time t . Show that the work done by the explosion is

$$\frac{1}{2} \cdot \frac{m_1 \cdot m_2}{m_1 + m_2} \cdot \frac{s^2}{t^2}$$

Ans. Let v_1 and v_2 be the velocities after explosion of the portion m_1 and m_2 respectively in opposite directions along the tube. Hence by the principle of conservation of momentum, we get

$$m_1 v_1 - m_2 v_2 = 0$$

$$\text{or } m_1 v_1 = m_2 v_2 \quad \text{--- (i)}$$

The distance apart between the portions after time t is

$$(v_1 + v_2)t = s \quad \text{--- (ii)}$$

From (i), we get, $\frac{v_1}{m_2} = \frac{v_2}{m_1} = \frac{v_1 + v_2}{m_1 + m_2} = \frac{s}{t(m_1 + m_2)} \quad [\text{by (ii)}]$

The work done by the explosion is the K.E. generated by it

$$= \frac{1}{2} (m_1 v_1^2 + m_2 v_2^2)$$

$$= \frac{1}{2} [m_1 m_2^2 + m_2 m_1^2] \left\{ \frac{s}{t(m_1 + m_2)} \right\}^2 \quad [\text{by (3)}]$$

$$= \frac{1}{2} m_1 m_2 [m_1 + m_2] \cdot \frac{s^2}{t^2 (m_1 + m_2)^2}$$

$$= \frac{1}{2} \cdot \frac{m_1 m_2}{m_1 + m_2} \cdot \frac{s^2}{t^2}$$

§ Problem 12:- For a rectilinear motion of a particle, if an impulse P changes its velocity from u to v and if E be the change of K.E., then show that

$$E = \frac{1}{2} P(u + v)$$

1. Newton's law for direct impact:

$$v_1 - v_2 = -e(u_1 - u_2) \quad (0 < e < 1)$$

2. For oblique impact

$$v_1 \cos \alpha_1 - v_2 \cos \alpha_2 = -e(u_1 \cos \alpha_1 - u_2 \cos \alpha_2)$$

3. Loss K.E for direct impact

$$\frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (1 - e^2) (u_2 - u_1)^2$$

4. Loss K.E for oblique impact $\frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (1 - e^2) (u_2 \cos \alpha_2 - u_1 \cos \alpha_1)^2$

5. Velocity of rebound = $e \times$ velocity of impact

$$\boxed{v = e u}$$

6. Height of rebound = $e \times$ height of impact

$$\Rightarrow \boxed{h = e^2 h}$$

7. Impulse of the sphere on the plane $(P) = m(1+e)\sqrt{2gh}$

8. Loss of K.E = $mgh(1 - e^2)$

9. Total distance travelled = $\frac{1+e^2}{1-e^2} h$

10. Time of flight = $\frac{1+e}{1-e} \sqrt{\frac{2h}{g}}$

Now with the help of ③ and ④ and proceeding exactly as in the case of direct impact

8 From what height must a heavy elastic ball be dropped on the floor so that rebounding on it it will reach a height of 8 m. ? $e = \frac{1}{2}$

Ans. Let the ball being dropped from a height of h , hit the floor with a velocity of u .

$$\text{Then } u^2 = 2gh$$

If v be the velocity of the ball just after impact with the floor.

$$v = eu$$

So, the height attained by the ball after first rebound

$$= \frac{v^2}{2g} = \frac{e^2 u^2}{2g} = \frac{e^2 2gh}{2g} = e^2 h = \left(\frac{1}{2}\right)^2 \cdot h = \frac{h}{4}$$

By the problem

$$\frac{h}{4} = 8$$

$$\Rightarrow h = 32$$

$\therefore$ It will reach a height of 32 m if it falls from 32 m.

3. A smooth sphere falling vertically from a height h impinges on a fixed smooth horizontal plane and rebounds to a height h_1 . Show that $h_1 = e^2 h$, where e is the co-efficient of restitution between the sphere and the plane.

Ans. Let u be the velocity of the sphere just before impact and v its velocity just after impact.

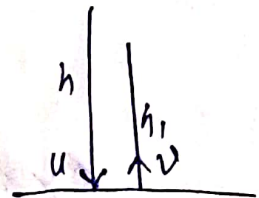
$$\text{Then } u^2 = 2gh \text{ and } v = eu, \quad v^2 = 2gh_1$$

$$\text{Now } v = eu$$

$$\Rightarrow e^2 u^2 = 2gh_1$$

$$\Rightarrow e^2 \times 2gh = 2gh_1$$

$$\Rightarrow \boxed{e^2 h = h_1}$$



§9 Prob. 3:- Two equal elastic spheres each of mass m move in the same direction with equal velocities u and collide directly. Show that they interchange their velocities.

Ans. Let v_1 and v_2 be the velocities of the spheres after impact. By Newton's law on impact,

$$v_2 - v_1 = e(u - u) = 0$$

$$\Rightarrow v_2 = v_1$$

Also, by the principle of conservation of linear momentum,

$$m v_1 + m v_2 = m u + m u$$

$$\Rightarrow v_1 = v_2 = u$$

$\therefore$ So, the velocities are interchanged.

§ Prob. 4:- Two elastic spheres impinge directly with equal and opposite velocities. Find the ratio of their masses so that one of these may come to rest after impact, the coefficient of restitution being e .

Ans. Let m_1 & m_2 be the masses of the spheres and u be their velocities (in opposite directions).

One of the spheres comes to rest after impact and let the velocity of the other sphere be v after impact.

Then by Newton's law on impact

$$0 - v = e(u + u) = 2eu$$

$$\therefore v = -2eu$$

By the principle of conservation of linear momentum,

$$m_1 v = m_1 u - m_2 u$$

$$\text{or } (1 + 2e)/m_1 u = m_2 u$$

$$\text{or } m_1 : m_2 = 1 : (1 + 2e)$$

Ex 10.5:- A sphere strikes directly on a second sphere at rest. If the co-efficient of restitution be e , then show that their velocities after impact are in the ratio $(1-e):(1+e)$.

Ans. Let u_1 be the velocity of the first sphere before impact and m be masses of both the spheres. Also, let v_1 and v_2 be their velocities after impact.

By Newton's experimental law

$$v_2 - v_1 = e(u_1 - 0) = eu_1 \quad \text{--- (1)}$$

Also, by the principle of conservation of linear momentum.

$$mu_1 = mv_2 + mv_1$$

$$\Rightarrow u_1 = v_2 + v_1 \quad \text{--- (2)}$$

$$\text{From (1) \& (2)} \quad \frac{v_2 - v_1}{v_2 + v_1} = \frac{eu_1}{u_1}$$

$$\Rightarrow \frac{v_2 - v_1}{v_2 + v_1} = \frac{e+1}{1-e}$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{1-e}{1+e}$$

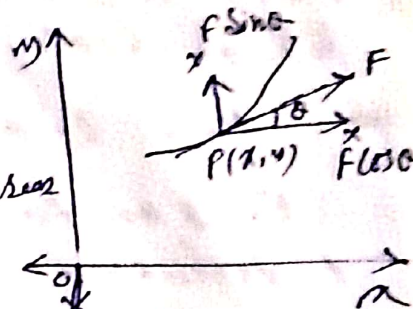
Motion in a plane

$\dot{x} = \frac{dx}{dt}$ = Velocity of the particle at P in the direction of x - increasing.

$\dot{y} = \frac{dy}{dt}$ = Velocity of the particle at P in the direction of y - increasing.

$\ddot{x} = \frac{d^2x}{dt^2}$ = acceleration of the particle at P in the direction of x - increasing.

$\ddot{y} = \frac{d^2y}{dt^2}$ = acceleration of the particle at P in the direction of y - increasing.



If F be the external force acting on a particle at P which makes an angle θ with the positive direction of x -axis.

Then the eqs of motion at P are

$$m\ddot{x} = F \cos \theta$$

$$m\ddot{y} = F \sin \theta$$

§ Relation between Angular Velocity & linear Velocity.

$$\omega = \frac{v}{r} \quad \left[v = \frac{\omega r}{1} \right]$$

for circle $r = r$.

$$\boxed{v = \omega r}$$

§ $v = \sqrt{\dot{x}^2 + \dot{y}^2}$

§ Path of a projectile in a vacuum:-

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

§ Angular Velocity: $\omega = \frac{d\theta}{dt}$

§ Polar Intrinsic co-ordinates (r, ψ) :

Velocity: $v = \frac{ds}{dt}$ (along tangent)

Tangential acceleration: $f_t = \frac{dv}{dt} = \frac{ds}{dt} = v \frac{dv}{ds}$

Normal " : $f_n = \frac{v^2}{r}$

8 Prob1. The position of a moving particle at any time t is given by $x = a \cos t$, $y = a \sin t$. Find its path, Velocity and acceleration.

Ans. Here $x = a \cos t$, $y = a \sin t$

The equation of the path is obtained by eliminating t between the expression for x and y .

$\therefore$ The equation of the path is

$$x^2 + y^2 = a^2$$

which represents a circle of radius a .

Again, $\dot{x} = -a \sin t$, $\dot{y} = a \cos t$

or v be the velocity,

$$v = \dot{x}^2 + \dot{y}^2 = a^2 \sin^2 t + a^2 \cos^2 t = a^2$$

$$\therefore v = a$$

and making an angle $\tan^{-1} \left(\frac{a \cos t}{-a \sin t} \right)$
 $= \tan^{-1} (\tan(\pi/2 + t))$
 $= \pi/2 + t$ with x -axis.

Also, $\ddot{x} = -a \cos t$, $\ddot{y} = -a \sin t$ a -axis.

$$v = \ddot{x}^2 + \ddot{y}^2 = a^2 (\cos^2 t + \sin^2 t) = a^2$$

$\therefore v = a$, in a direction making an angle $\tan^{-1} \left(\frac{-a \sin t}{-a \cos t} \right) = \tan^{-1} (\tan t)$
 $= t$ with x -axis.

8 Prob2:- A particle moves in a plane, its velocity parallel to the axes of x and y are $u + \omega_1 y$ and $v + \omega_2 x$ resp. where u, v, ω_1, ω_2 are constant. Show that the path of the particle is a conic section.

Ans. $\frac{dx}{dt} = u + \omega_1 y$ & $\frac{dy}{dt} = v + \omega_2 x$

$$\therefore \frac{dy}{dx} = \frac{v + \omega_2 x}{u + \omega_1 y}$$

$$\Rightarrow \frac{dy}{u + \omega_1 y} = \frac{v + \omega_2 x}{\omega_1} \Rightarrow \frac{1}{\omega_1} \log(u + \omega_1 y) = \frac{1}{\omega_2} \log(v + \omega_2 x) + \log C$$

$$\frac{dy}{dx} = \frac{v + R_2 x}{u + R_1 y}$$

$$\Rightarrow (u + R_1 y) dy = (v + R_2 x) dx$$

$$\Rightarrow u y + R_1 y^2/2 = v x + R_2 x^2/2 + C$$

$$\Rightarrow 2(u y - v x) + (R_1 y^2 - R_2 x^2) = 2C$$

8 Prob 3: A particle is moving with a constant velocity parallel to the axis of y and velocity proportional to y parallel to the axis of x . Show that the path of the particle is a parabola.

Ans. Let (x, y) be the position of the particle at any time t .

Then $\frac{dx}{dt} = Ay$ and $\frac{dy}{dt} = \mu$ [where A & μ are const]

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\mu}{Ay}$$

$$\Rightarrow Ay dy = \mu dx$$

$$\Rightarrow y^2/2 = \mu x + C \quad [\text{Integrating}]$$

$$\Rightarrow y^2 = 2\mu x + 2C$$

$$\Rightarrow y^2 = \frac{2\mu}{A} \left(x + \frac{C}{\mu} \right)$$

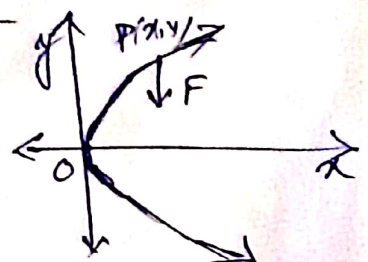
which represents a parabola whose axis is along the x -axis, vertex at $(-C/\mu, 0)$ and latus rectum $\frac{2\mu}{A}$.

8 Prob 4: A particle describes a parabolic path moving under a force which is always perpendicular to its axis. Find the law of force and the velocity of the particle at any point in the orbit.

Ans. without any loss of generality, we take the equation of the parabola as

$$y^2 = 4ax \quad \text{--- (1)}$$

whose vertex is at origin and axis along x -axis



Equations of motion of the ~~particle~~ particles are

$$\ddot{x} = 0 \quad \text{--- (1)}$$

$$\ddot{y} = -F \quad \text{--- (2)}$$

F being the force per unit mass directed towards the axis of parabola, which is x -axis from (2) on integration.

$$\ddot{x} = c = \text{constant}$$

d. w. r. to t

$$2m\dot{x} = 4aq\dot{x} = 4ac$$

$$\Rightarrow \dot{x} = \frac{2ac}{m}$$

d. w. r. to t

$$\ddot{y} = - \frac{2ac}{m\dot{x}} \cdot \dot{y}$$

$$= - \frac{2ac}{m\dot{x}} \cdot \frac{2ac}{m}$$

$$a \rightarrow F = - \frac{4a^2c^2}{m^2} \quad [by (1)]$$

$$\Rightarrow F = \frac{4a^2c^2}{m^2}$$

$\Rightarrow$ for $\frac{1}{m^2}$

This is the law of force.

If v be the velocity at any point (x, y)

$$v^2 = \dot{x}^2 + \dot{y}^2 = c^2 + \frac{4a^2c^2}{m^2}$$

$$= \frac{c^2}{m^2} (m^2 + 4a^2)$$

$$= \frac{c^2}{4a^2} (4a^2 + 4a^2)$$

$$or, v^2 = \frac{c^2}{4a^2} (x + a)$$

$$\Rightarrow v = c \sqrt{\frac{x+a}{4a^2}}$$

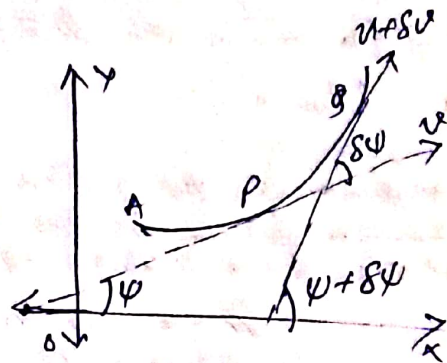


§ Motion in a plane: Tangent & Normal Acceleration

Velocity at P = $\frac{ds}{dt}$

Acceleration along the tangent at

$$P = \frac{dv}{dt} = \frac{dv}{ds} \cdot \frac{ds}{dt} = v \cdot \frac{dv}{ds}$$



Acceleration along inward normal at P

$$= \frac{v^2}{\rho} \quad \left[\rho = \frac{ds}{d\psi} \right]$$

$$\therefore \dot{\psi} = \frac{d\psi}{dt} = \frac{d\psi}{ds} \cdot \frac{ds}{dt} = \frac{1}{\rho} \Rightarrow \boxed{v = \rho \dot{\psi}}$$

§ Prob 1:- For a particle moving along a plane curve, if the tangential and normal acceleration are equal in magnitude. Show that the Velocity varies as e^{ψ} .

Ans. Since the tangential component of acceleration is equal to the normal component of acceleration.

$$v \cdot \frac{dv}{ds} = \frac{v^2}{\rho} = v \cdot \frac{d\psi}{ds}$$

$$\Rightarrow \frac{dv}{v} = d\psi$$

$$\Rightarrow \log v = \psi + \log K$$

$$\Rightarrow \log v - \log K = \psi$$

$$\Rightarrow v = K \cdot e^{\psi}$$

$$\Rightarrow v \propto e^{\psi}$$

§ Prob 2:- A particle moves in the curve $y = a \log x (x > 0)$ in such a way that the tangent to the curve rotates uniformly. Prove that the resultant acceleration varies as the square of the radius of curvature.

$$\text{Ans. } \tan \phi = \frac{dy}{dx} = a \cdot \frac{du/dx \cdot \tan \theta}{du(dx)} \cdot \frac{1}{u}$$

$$\Rightarrow \tan \phi = \tan \theta$$

$$\Rightarrow \phi = \theta \quad \text{--- (1)}$$

As the tangent rotates uniformly,

$$\dot{\phi} = \dot{\theta} = \text{constant.}$$

$$v = \frac{ds}{dt} = \frac{ds}{dx} \cdot \frac{dx}{dt} \cdot \frac{dy}{dx}$$

$$= \frac{ds}{dx} \cdot \frac{dx}{dt} \cdot \frac{dy}{dx}$$

$$= \sec \phi \cdot a \cdot \dot{\theta}$$

$$\left[\begin{array}{l} \frac{dy}{dx} = \tan \theta \\ \frac{dx}{ds} = \cos \theta, \quad \frac{dy}{ds} = \sin \theta \end{array} \right]$$

$$f_t = \frac{dv}{dt} = a \dot{\theta} \sec \phi \tan \phi \cdot \frac{dy}{dx}$$

$$= a \dot{\theta} \sec \phi \tan \phi$$

$$f_n = \frac{v^2}{\rho} = \frac{v^2}{\frac{ds}{dx}} = \frac{v^2}{\frac{ds}{dx} \cdot \frac{dx}{dy}} = \frac{\sec^2 \phi a^2 \dot{\theta}^2}{\sec \phi \cdot a}$$

$$= a \dot{\theta}^2 \sec \phi$$

If f be the acceleration

$$f = \sqrt{f_t^2 + f_n^2} = \sqrt{a^2 \dot{\theta}^4 \sec^2 \phi \sec^2 \phi} = a \dot{\theta}^2 \sec^2 \phi$$

$$= \frac{\dot{\theta}^2}{a} \rho$$

$$\boxed{a/f \propto \rho}$$

8 Prove that the acceleration of a particle moving in a curve with uniform speed is $\frac{v^2}{\rho}$.

Ans. Since v is const. f_t = tangential component of acceleration

$$\dot{\phi} = \frac{d\theta}{dt} = \frac{ds}{ds} \cdot \frac{d\theta}{ds} = v/\rho$$

$$a \dot{\theta} = v/\rho$$

$$\boxed{f = \frac{v^2}{\rho}} \quad f_n = \frac{v^2}{\rho} = \frac{\rho^2 \dot{\phi}^2}{\rho} = \rho \dot{\phi}^2$$

Motion in a plane Polar Co-ordinates

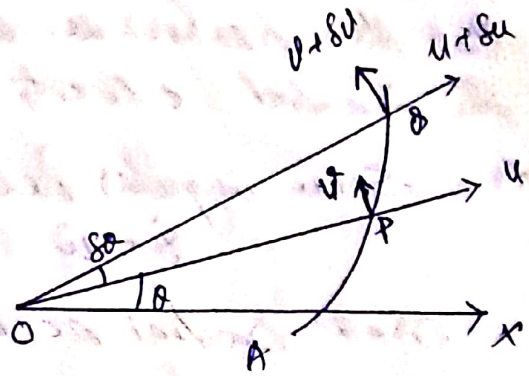
§ Radial Velocity = $\dot{r}$

§ Cross Radial Velocity = $r\dot{\theta}$

§ Radial Acceleration = $\ddot{r} - r\dot{\theta}^2$

§ Cross Radial Acceleration

$$u = \frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta})$$



§ Problem:- If the radial velocity of a particle is proportional to its transverse velocity. Find the equation of the path of the particle in polar co-ordinates.

Ans. Here Radial Velocity is proportional to the transverse velocity.

So, $\dot{r} = k r \dot{\theta}$, k being constant of proportionality

$$\Rightarrow \frac{dr}{r} = k d\theta$$

$$\Rightarrow \ln r = k\theta + \ln c$$

$$\Rightarrow \ln r = k\theta + \ln c$$

$$\Rightarrow r = c \cdot e^{k\theta}$$

$$\Rightarrow r \propto e^{k\theta}$$

§ Problem:- A particle describes the curve $r = ae^{k\theta}$ with constant angular velocity. Show that the radial acc is zero.

Ans. Since the angular velocity is const.

$$\therefore \dot{\theta} = \omega = \text{constant}$$

$$\text{Radial Acceleration} = \ddot{r} - r\dot{\theta}^2 = a k^2 e^{k\theta} - a e^{k\theta} \omega^2 = 0$$

Q. Prob 31:- If the Angular Velocity of a moving point about a fixed origin be const. Show that the transverse acceleration varies as the radial velocity.

Ans. Since angular velocity is const.

$$\dot{\theta} = \omega \text{ (const.)}$$

$$\text{transverse acceleration} = \frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta})$$

$$= \frac{1}{r} \frac{d}{dt} (r^2 \omega)$$

$$= \frac{1}{r} \cdot \omega \cdot 2 \cdot r \cdot \dot{r}$$

$$= 2\omega \dot{r}$$

$$\Rightarrow \boxed{f_N \propto \dot{r}}$$

Q. Prob 4:- A particle describes the curve $r = ae^{\theta}$ with constant angular velocity. Show that the transverse acceleration varies as the distance from the pole.

Ans. Since angular velocity is const.

$$\dot{\theta} = \omega \text{ (const.)}$$

$$\text{transverse acceleration} = \frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta})$$

$$= \frac{1}{r} \frac{d}{dt} (r^2 \omega)$$

$$= \frac{1}{r} \cdot \omega \cdot 2 \cdot r \cdot \dot{r}$$

$$= 2\omega \dot{r}$$

$$= 2\omega \cdot a \cdot e^{\theta} \cdot \dot{\theta}$$

$$= 2\omega \cdot a \cdot e^{\theta} \cdot \omega$$

$$= 2\omega^2 r$$

$$\Rightarrow \boxed{f_N \propto r}$$

3. Prob. A particle describes the parabola $y = ax^2$ such that the cross radial component of the velocity is constant. Show that $\dot{\theta}$ is const.

The given that equation of the path is $y = ax^2$

Cross Radial Component of Velocity (V_θ)

$$= K \text{ (const.)} \Rightarrow r \dot{\theta} = K \Rightarrow r = \frac{K}{\dot{\theta}}$$

$$\text{Now } \ddot{r} = \frac{d}{dt}(\dot{r}) = \frac{d}{dt} \left(a \frac{d}{dx} \cdot \tan \theta \cdot \frac{\partial}{\partial x} \right)$$

$$= \frac{d}{dt} \left(a \frac{d}{dx} \cdot \tan \theta \cdot \frac{K}{r} \right)$$

$$= \frac{d}{dt} \left(a \frac{d}{dx} \cdot \tan \theta \cdot K \cdot \frac{1}{r} \right)$$

$$= \frac{d}{dt} (K \sin \theta \cdot \tan \theta)$$

$$= K \left(\frac{d}{dt} \cos \theta \cdot \tan \theta + \sin \theta \cdot \frac{d}{dt} \tan \theta \right)$$

$$= K \left(-\sin \theta + \tan \theta \cdot \sin \theta \right) \cdot \frac{1}{2v}$$

$$= K \sin \theta$$

$$\therefore \frac{d}{dt} \left(\frac{K}{\dot{\theta}} \right) = \frac{d}{dt} \left(\frac{K}{\dot{\theta}} \right) = a \frac{d}{dx} \cdot \tan \theta \cdot \frac{\partial}{\partial x}$$

$$= a \tan \theta \cdot \dot{\theta}$$

$$= K \tan \theta$$

$$\ddot{r} = \frac{d}{dt}(\dot{r}) = \frac{d}{dt} (K \tan \theta)$$

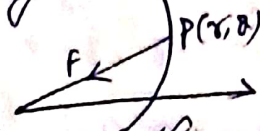
$$= K \sec^2 \theta \cdot \dot{\theta}$$

$$= K/2 \cdot K/4 \cdot 1/a = \frac{K^2}{8a} = \text{const.}$$

Central Orbit

§ If a force F acting on a particle of mass m be such that it is always directed towards or away from a fixed point and

1) the magnitude of the force F at any time is a function of the distance r of the particle from the fixed point, then the force is said to be a central force. The path described by a particle under the action of a central force is called a central orbit.



§ The fixed point, towards or away from which the force always acts, is called the centre of force.

§ Differential equation of the path in polar-coordinates (r, θ) .

$$\frac{d^2 u}{d\theta^2} + u = \frac{F}{h^2 u^3} \quad \text{where } F \text{ being the force per unit mass and } u = \frac{1}{r}.$$

§ Differential equation of the path in pedal form:-

$$\left[\frac{h^2}{p^3} - \frac{dp}{dr} = F \right]$$

§ In a central orbit, the areal area traced out by the radius vector drawn from the centre of force to the position of the moving particle increase at a constant rate given by $\frac{h^2}{2}$, where $h = r^2 \dot{\theta}$.

§ In a central orbit, the moment of momentum or the angular momentum remains constant.

$$\vec{v} = \dot{r}\vec{e}_r + (r\dot{\theta})\vec{e}_\theta \quad \text{1) } h = p v = r^2 \dot{\theta} \quad \text{2) } \vec{v} = h \left(\frac{1}{r^2} \vec{e}_r + \left(\frac{1}{r} \frac{d}{d\theta} \right) \vec{e}_\theta \right)$$

§ At an apse $\frac{dy}{d\theta} = 0$, $p = r$, $\phi = \pi/2$.

$$\left[v_0 = \mu \alpha \right] \quad \left[\frac{h^2}{r} = \frac{2\mu}{r} \right]$$

§ Apse:- An apse on a central orbit is a point at which the distance r from the centre is either maximum or minimum.

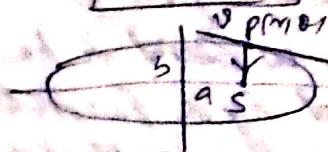
8 Circular orbit: The centre of force being on the circumference.

1. $f \propto r^5$ 2. $v \propto r^2$



9 Elliptic orbit: Force directed to a focus.

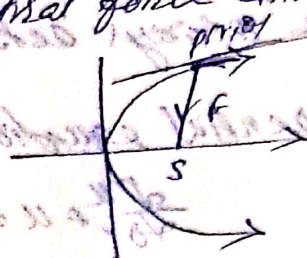
1. $f \propto r^{-2}$ 2. $v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$



3. $T = \frac{\text{Area of the ellipse}}{h/2} = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$

10 Parabolic orbit under a central force directed towards the focus:-

1. $f \propto r^{-2}$ 2. $v = \frac{h}{\sqrt{a}r}$
3. $v \propto \frac{1}{\sqrt{r}}$



11 Inverse square law:-

$F = \mu/r^2$, v be the initial velocity at a distance R .

i) The orbit is an ellipse if $v^2 < \frac{2\mu}{R}$

ii) The orbit is a parabola if $v^2 = \frac{2\mu}{R}$

iii) The orbit is a hyperbola if $v^2 > \frac{2\mu}{R}$

12 Velocity at any point:-

i) for an ellipse: $v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$, where $h^2 = \mu a (1 - e^2) = \mu a (1 - e^2)$

ii) for a hyperbola: $v^2 = \mu \left(\frac{2}{r} + \frac{1}{a} \right)$

iii) parabola: $v^2 = \frac{2\mu}{r}$

1. A particle describes the path $r^4 = a^4 \cos 4\theta$ under a force which is always directed to the pole. Find the law of force.

Ans. $r^4 = a^4 \cos 4\theta$
 $\Rightarrow 4 \log r = 4 \log a + \log \cos 4\theta$
 $\Rightarrow 4 \cdot \frac{1}{r} \cdot \frac{dr}{d\theta} = \frac{1}{\cos 4\theta} \cdot (-\sin 4\theta \cdot 4)$
 $\Rightarrow \cot \phi = -\tan 4\theta$
 $\Rightarrow \cot \phi = \cot (4\theta + \pi/2)$
 $\Rightarrow \phi = 4\theta + \pi/2$

$p = r \sin \phi$
 $= r \sin (4\theta + \pi/2)$
 $p = r \cos 4\theta = r \cdot \frac{a^4}{r^4} = \frac{a^4}{r^3}$
 $\frac{dp}{dr} = 5 \cdot r^4 \cdot \frac{1}{a^4}$
 $\therefore f = \frac{h^2}{p^3} \cdot \frac{dp}{dr} = h^2 \cdot \left(\frac{a^4}{r^5}\right)^3 \cdot \frac{5 \cdot r^4}{a^4}$
 $\Rightarrow f = h^2 \cdot \frac{5 \cdot a^8}{r^{11}} \Rightarrow \boxed{f \propto \frac{1}{r^{11}}}$

2. A particle describes the equiangular spiral $r = a e^{c\theta}$ under a force directed towards the pole. Find the law of force and velocity in terms of r .

Ans. $r = a e^{c\theta}$
 $\Rightarrow a u e^{c\theta} = r, \quad \frac{dr}{d\theta} = u$
 $\text{Ave. } e^{c\theta} \cdot \frac{du}{d\theta} + a u \cdot e^{c\theta} = 0$
 $\Rightarrow \frac{du}{d\theta} + u \cot \theta = 0 \quad a e^{c\theta} \neq 0$
 $\text{Ave. } u \cdot r \neq 0$

$\frac{du}{d\theta} + \frac{du}{d\theta} \cot \theta = 0$
 $\Rightarrow \frac{du}{d\theta} = -u \cot \theta$

Now $\frac{du}{d\theta} + u = \frac{F}{h^2 u} \Rightarrow u \cot \theta + u = \frac{F}{h^2 u} \Rightarrow F = h^2 u^3 \cot \theta$
 $= \frac{h^2 \cot \theta}{r^3}$
 $\Rightarrow \boxed{f \propto \frac{1}{r^3}}$